

From Manual Scaling to Intelligent Automation

How Cost Visibility Saved Our Team

Esmira • CAST AI

A QUESTION FOR THE ROOM

**What percentage of your
cloud compute do you think
is actually being used
right now?**

Take a guess. 50%? 60%?

THE REAL NUMBER

**92 to 95 cents of every dollar
is paying for capacity
nobody is using.**

8%

average CPU utilization

5%

average GPU utilization

Across 23,000+ Kubernetes clusters analyzed · Hold onto this number — I'll come back to it.

WHO I AM

DevOps leadership at a U.S. federal space organization

- Building and launching satellites — almost every other month
- No “we'll fix it later” when what you ship is in orbit
- Technical debt piles up fast when you're always moving to the next launch

This talk: the six months I was told to cut 20% of our cloud cost — or cut engineers.

THE MANDATE

“Cut cloud cost 20% in six months — or cut engineers.”

That came down from leadership, out of nowhere from where I sat. My first thought wasn't “how do I optimize the bill.” It was “who do I let go.”

WHY THAT WASN'T A SOLUTION

Two orders that couldn't both be true.

Already short-staffed

Not comfortably staffed. Short.

Still shipping every other month

Satellite schedules don't move for a budget review

Cut engineers?

Same work, fewer people, on top of existing technical debt

THE REAL PROBLEM

Not “we use too much.”

“We provision far more than we use.”

Every team had padded its resource requests, just to be safe. Every workload carried headroom nobody had gone back to check.

The moment I saw the gap clearly, the plan changed: this wasn't a “who do we cut” problem. It was a “why are we paying for compute we don't use” problem.

Reallocate, don't cut.



Instead of this

- Cut engineers to hit the budget target
- Ask stretched engineers to manually tune every service
- Add a new project on top of existing load



We did this

- Assigned resources based on actual need, not “just in case”
- Distributed workload smarter across the team and infra
- Fit the fix inside capacity we already had

The system stayed efficient on its own.

1

Bin-pack

Pack workloads more intelligently onto the infrastructure we already had

2

Automate

Make it stay true automatically — new services, changing traffic, no manual check-ins

3

No dependency on memory

Waste doesn't creep back in because someone forgot to look

THE RESULT

We hit 20%. Without letting a single engineer go.

The team stayed exactly the size it needed to keep building satellites on schedule.

Leadership got the number they asked for — just not the way they proposed getting there.

ZOOMING OUT

This isn't unique to satellites.

Every team that treats infrastructure spend as fixed overhead is quietly volunteering its own headcount as the adjustment variable. I know, because I was handed exactly that choice.

5%

average GPU utilization

*about to matter a lot more as AI workloads eat bigger
shares of every budget*

Three things

01

Don't accept the framing

When a cost mandate lands, cutting people isn't the only lever — check utilization before you agree to that trade-off

02

Visibility before optimization

You cannot right-size what you cannot see. I couldn't fix anything until I saw the gap

03

Automate the boring part

Bin-packing and right-sizing are maintenance, not judgment calls — let automation hold the line

CLOSING

**Whatever that gap is on
your team — that's not a
cost problem.**

*It's the budget for your next engineer — or the job of the one sitting next to you — already
hiding inside your current cloud bill.*

Go find it before someone hands you the same ultimatum I got.